

Practica 1: Diseño de controlador para un sistema de segundo orden

Ecuaciones principales

$$V_e(t) = R i_1(t) + L \frac{d[i_1(t) - i_2(t)]}{dt} + R [i_1(t) - i_2(t)]$$
$$L \frac{d[i_1(t) - i_2(t)]}{dt} + R [i_1(t) - i_2(t)] = R i_2(t) + R \int i_2(t) dt + \frac{1}{C} \int i_2(t) dt$$
$$V_s(t) = R i_2(t) + \frac{1}{C} \int i_2(t) dt$$

Transformada de Laplace

$$V_e(s) = R I_1(s) + L s [I_1(s) - I_2(s)] + R [I_1(s) - I_2(s)]$$
$$L s [I_1(s) - I_2(s)] + R [I_1(s) - I_2(s)] = R I_2(s) + R I_2(s) + \frac{I_2(s)}{C s}$$
$$V_s(s) = R I_2(s) + \frac{I_2(s)}{C s}$$

Procedimiento Algebraico

$$V_e(s) = (L s + 2R) I_1(s) - (L s + R) I_2(s)$$

$$I_1(s) = \frac{1}{L s + R} \left(L s + \frac{1}{C s} + 3R \right) I_2(s) = \frac{C L s^2 + 3R C s + 1}{C L s^2 + C R s} I_2(s)$$

$$V_s(s) = \left(R + \frac{1}{C s} \right) I_2(s) = \frac{C R s + 1}{C s} I_2(s)$$

$$V_e(s) = \left[(L s + 2R) \frac{C L s^2 + 3R C s + 1}{C L s^2 + C R s} - (L s + R) \right] I_2(s)$$

Función de transferencia

$$\frac{V_s(s)}{V_e(s)} = \frac{\frac{C R s + 1}{C s} I_2(s)}{\frac{3C L R s^2 + (3C R^2 + L) s + 2R}{C L s^2 + C R s} I_2(s)} = \frac{(C R s + 1)(L s + 2R)}{3C L R s^2 + (3C R^2 + L) s + 2R}$$

Error en estado estacionario

$$e_{ss} = \left(1 - \frac{V_s}{V_e} \right) = \left(1 - \frac{C L R s^2 + (C R^2 + L) s + R}{3C L R s^2 + (3C R^2 + L) s + 2R} \right) = 1 - \frac{1}{2} = 0.5$$

estabilidad

$$R = 100 \text{ k}\Omega \quad a = 3C L R = 0.03199$$
$$C = 470 \text{ nF} \quad b = 5C R^2 + L = 2350000$$
$$L = 390 \text{ mH} \quad c = 2R = 200000$$

$$s_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2ac}$$

$$s_1 = -8.511 \times 10^{-3}$$

$$s_2 = -427350427.3$$

Respuesta: Subamortiguado

Bosquejo:

